Test Series Code

11

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | **DROPPER** | **M. MARKS: 720**

DATE: 05/04/2026

ANSWER KEY

1.	(2)	46.	(3)	91.	(2)	136.	(2)
2.	(1)	47.	(4)	92.	(3)	137.	(3)
3.	(4)	48.	(1)	93.	(1)	138.	(3)
4.	(2)	49.	(4)	94.	(2)	139.	(4)
5.	(1)	50.	(2)	95.	(4)	140.	(3)
6.	(2)	51.	(1)	96.	(3)	141.	(3)
7.	(3)	52.	(2)	97.	(2)	142.	(3)
8.	(2)	53.	(3)	98.	(1)	143.	(3)
9.	(1)	54.	(1)	99.	(4)	144.	(2)
10.	(2)	55.	(3)	100.	(3)	145.	(1)
11.	(2)	56.	(1)	101.	(4)	146.	(4)
12.	(1)	57.	(1)	102.	(1)	147.	(3)
13.	(4)	58.	(1)	103.	(3)	148.	(1)
14.	(1)	59.	(2)	104.	(2)	149.	(3)
15.	(2)	60.	(1)	105.	(2)	150.	(2)
16.	(1)	61.	(1)	106.	(3)	151.	(2)
17.	(2)	62.	(1)	107.	(2)	152.	(1)
18.	(3)	63.	(2)	108.	(2)	153.	(2)
19.	(1)	64.	(4)	109.	(1)	154.	(2)
20.	(2)	65.	(2)	110.	(4)	155.	(2)
21.	(4)	66.	(3)	111.	(3)	156.	(4)
22.	(3)	67.	(2)	112.	(2)	157.	(2)
23.	(1)	68.	(3)	113.	(1)	158.	(3)
24.	(2)	69.	(2)	114.	(4)	159.	(4)
25.	(4)	70.	(4)	115.	(3)	160.	(3)
26.	(1)	71.	(2)	116.	(2)	161.	(4)
27.	(3)	72.	(4)	117.	(2)	162.	(1)
28.	(3)	73.	(4)	118.	(1)	163.	(4)
29.	(3)	74.	(4)	119.	(3)	164.	(2)
30.	(4)	75.	(2)	120.	(2)	165.	(3)
31.	(1)	76.	(1)	121.	(3)	166.	(2)
32.	(2)	77.	(2)	122.	(4)	167.	(2)
33.	(3)	78.	(2)	123.	(1)	168.	(4)
34.	(1)	79.	(3)	124.	(3)	169.	(3)
35.	(2)	80.	(3)	125.	(2)	170.	(3)
36.	(2)	81.	(2)	126.	(3)	171.	(1)
37.	(4)	82.	(3)	127.	(4)	172.	(4)
38.	(1)	83.	(3)	128.	(4)	173.	(2)
39.	(3)	84.	(3)	129.	(1)	174.	(3)
40.	(2)	85.	(3)	130.	(3)	175.	(2)
41.	(1)	86.	(4)	131.	(2)	176.	(2)
42.	(3)	87.	(2)	132.	(1)	177.	(2)
43.	(2)	88.	(1)	133.	(3)	178.	(3)
44.	(3)	89.	(1)	134.	(1)	179.	(4)
45.	(3)	90.	(2)	135.	(4)	180.	(2)

[Contd...

1. (2)

$\frac{1}{F_{eq}}=\frac{1}{F_{rem}}+\frac{1}{f_3}\Rightarrow\frac{1}{20}=\frac{1}{15}+\frac{1}{f_3}$.

$\frac{1}{f_3}=\frac{1}{20}-\frac{1}{15}=\frac{3-4}{60}=-\frac{1}{60}$.

Negative sign implies diverging lens.

(NCERT Ed. 2025-26, 12th, Page No. 237)

2. (1)

Loss in PE = Gain in Rotational KE.

$mg(\frac{L}{2})=\frac{1}{2}(\frac{mL^2}{3})\omega^2\Rightarrow\omega=\sqrt{\frac{3g}{L}}$

$v=L\omega=\sqrt{3gL}=\sqrt{3\times10\times1.2}=\sqrt{36}=6$ m/s.

(NCERT Ed. 2025-26, 11th, Page No. 122)

3. (4)

$\tau=\frac{\Delta L}{\Delta t}=\frac{6L_0-2L_0}{2}=\frac{4L_0}{2}=2L_0$.

(NCERT Ed. 2025-26, 11th, Page No. 121)

4. (2)

According to the Kinetic Theory of Gases, the average kinetic energy (*KE*) of gas molecules is directly proportional to the absolute temperature (*T*) of the gas, given by the relation: $KE_{avg}=\frac{3}{2}kT$.

Therefore, as temperature increases, the speed and kinetic energy of the gas molecules also increase.

(NCERT Ed. 2025-26, 11th, Page No. 231)

5. (1)

The electric field at a distance *R* from an infinite wire with charge density $+\lambda$ is: $E_w=\frac{\lambda}{2\pi\epsilon_0 R}$.

At *C*, this field points towards the left. The electric field at the center of a semicircular arc with charge density λ' is $E=\frac{2k\lambda'}{R}=\frac{\lambda'}{2\pi\epsilon_0 R}$.

Substituting $\lambda'=2\lambda$: $E_a=\frac{2\lambda}{2\pi\epsilon_0 R}=\frac{\lambda}{\pi\epsilon_0 R}$.

At *C*, this points right.

$E_{net}=E_a-E_w=\frac{2\lambda-\lambda}{2\pi\epsilon_0 R}=\frac{\lambda}{2\pi\epsilon_0 R}$ towards the right.

(NCERT Ed. 2025-26, 12th, Page No. 34)

6. (2)

Elastic Potential Energy

$U=\frac{1}{2}\times\text{Load}\times\text{Extension}=\frac{1}{2}\times200\times(4\times10^{-3})$

$=0.4$ J.

(NCERT Ed. 2025-26, 11th, Page No. 181)

7. (3)

For an electron, $\lambda=\frac{12.27}{\sqrt{V}}$ Å.

$\lambda=\frac{12.27}{\sqrt{100}}=1.227\text{Å}\approx0.123$ nm.

(NCERT Ed. 2025-26, 12th, Page No. 286)

8. (2)

Radius $r=\frac{p}{qB}$. Since *p* and *B* are constant, $r\propto\frac{1}{q}$.

$\frac{r_d}{r_\alpha}=\frac{q_\alpha}{q_d}=\frac{2e}{e}=\frac{2}{1}$.

(NCERT Ed. 2025-26, 12th, Page No. 112)

9. (1)

$W=\int_{y_1}^{y_2}Fdy=\int_0^2(3y+4)dy=\left[\frac{3y^2}{2}+4y\right]_0^2$

$=\left(\frac{3(4)}{2}+4(2)\right)-0=14$ J.

(NCERT Ed. 2025-26, 11th, Page No. 95)

10. (2)

For the block to not fall, Friction ≥ Weight. $\mu(m\omega^2 r)\geq mg$. $\omega_{min}=\sqrt{\frac{g}{\mu r}}=\sqrt{\frac{10}{0.2\times2}}=5$ rad/s.

(NCERT Ed. 2025-26, 11th, Page No. 78)

11. (2)

Let $k=\frac{I_{max}}{I_{min}}$. $\frac{I_{max}-I_{min}}{I_{max}+I_{min}}=\frac{k-1}{k+1}$.

$0.8=\frac{k-1}{k+1}\Rightarrow0.8(k+1)=k-1$

$\Rightarrow0.8k+0.8=k-1\Rightarrow k=9$.

(NCERT Ed. 2025-26, 12th, Page No. 264)

12. (1)

True weight $W=\sqrt{W_1W_2}=\sqrt{9\times25}=\sqrt{225}$

$=15$ kg.

(NCERT Ed. 2025-26, 11th, Page No. 125)

13. (4)

$v_{sep}=e(v_{app})\Rightarrow v_B-\left(\frac{-u}{3}\right)=1(u)\Rightarrow v_B=\frac{2u}{3}$.

Conservation of momentum:

$m_Au=m_B\left(\frac{2u}{3}\right)-m_A\left(\frac{u}{3}\right)\Rightarrow\frac{4}{3}m_Au=\frac{2}{3}m_Bu$

$\Rightarrow2m_A=m_B\Rightarrow\frac{m_B}{m_A}=2$.

(NCERT Ed. 2025-26, 11th, Page No. 101)

[Contd...

14. (1)

Let speed of B be u. Time for B to catch C: $t = \frac{d}{u-10}$. $x_C - x_A = 3d \Rightarrow (2d + 10t) - 5t = 3d$

$\Rightarrow 5t = d \Rightarrow t = \frac{d}{5}$.

$\frac{d}{5} = \frac{d}{u-10} \Rightarrow u - 10 = 5 \Rightarrow u = 15$ m/s.

(NCERT Ed. 2025-26, 11th, Page No. 41)

15. (2)

Since $V_R = V_L = V_C$, the circuit is in resonance. $V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{200}{\sqrt{2}} = 100\sqrt{2}$ V.

(NCERT Ed. 2025-26, 12th, Page No. 189)

16. (1)

Statement 1 is true (confirms null point lies on wire). Statement 2 is false (dragging alters wire resistance).

(NCERT Ed. 2025-26, 12th, Page No. 102)

17. (2)

$[c] = [t] = [T] \rightarrow (P)$.

$[a][T] = [LT^{-1}] \Rightarrow [a] = [LT^{-2}] \rightarrow (Q)$.

$\frac{[b]}{[T]} = [LT^{-1}] \Rightarrow [b] = [L] \rightarrow (R)$.

(NCERT Ed. 2025-26, 11th, Page No. 21)

18. (3)

$|\vec{v}_{rel}|^2 = |\vec{v}_{truck}|^2 + |\vec{v}_{cyclist}|^2 \Rightarrow 13^2 = v_{truck}^2 + 5^2$

$\Rightarrow 169 = v_{truck}^2 + 25 \Rightarrow v_{truck} = 12$ km/h.

(NCERT Ed. 2025-26, 11th, Page No. 62)

19. (1)

$x = x_0 = 2$ ($\because a_x = 0$)

$y = y_0 + u_y t + \frac{1}{2} a_y t^2 = 5 + 0 + \frac{1}{2}(6)(2)^2 = 17$.

Coordinates (2,17).

(NCERT Ed. 2025-26, 11th, Page No. 61)

20. (2)

Assuming $x = A \sin \omega t$

For $0 \rightarrow A/2$, $\omega t_1 = \frac{\pi}{6} \Rightarrow t_1 = \frac{T}{12}$.

For $0 \rightarrow A$, time is $\frac{T}{4}$.

$t_2 = \frac{T}{4} - \frac{T}{12} = \frac{T}{6}$. $t_2 = 2t_1$.

(NCERT Ed. 2025-26, 11th, Page No. 261)

21. (4)

$\omega = 100\pi$. $f = \frac{\omega}{2\pi} = 50$ Hz.

(NCERT Ed. 2025-26, 11th, Page No. 278)

22. (3)

$f_B = 300 \pm 5 \Rightarrow 305$ or 295. Loading wax decreases f_B . If 305 $\rightarrow$ 302, beat diff $\approx$ 2 (matches). If 295 $\rightarrow$ 292, beat diff increases. So 305 Hz.

(NCERT Ed. 2025-26, 11th, Page No. 286)

23. (1)

$f_{avg} = \frac{n_1 f_1 + n_2 f_2}{n_1 + n_2} = \frac{0.5(5) + 1(3)}{1.5} = \frac{5.5}{1.5} \approx 3.67$.

(NCERT Ed. 2025-26, 11th, Page No. 234)

24. (2)

$\frac{hc}{\lambda} \geq E \Rightarrow \lambda \leq \frac{hc}{E}$

$\lambda_{max} = \frac{hc}{E} = \frac{1240}{3.1} = 400$ nm.

(NCERT Ed. 2025-26, 12th, Page No. 344)

25. (4)

$k = \frac{80}{0.2} = 400$ N/m.

$T = 2\pi\sqrt{\frac{0.01}{400}} = \frac{2\pi}{200} \approx 0.0314$ s.

(NCERT Ed. 2025-26, 11th, Page No. 261)

26. (1)

(ii) AND, (iv) NAND, (iii) NOT.

(NCERT Ed. 2025-26, 12th, Page No. 348)

27. (3)

$v_{avg} = \frac{3v}{2}$. $\epsilon = B\left(\frac{3v}{2}\right)l$.

(NCERT Ed. 2025-26, 12th, Page No. 163)

28. (3)

Symmetric configuration cancels out potential energies of sides and face diagonal. Due to body diagonal pairs.

$U = \frac{4\,kq^2}{\sqrt{3}a} = 120$ J

(NCERT Ed. 2025-26, 12th, Page No. 56)

[Contd...

29. (3)

It has the exact orbital velocity to maintain the same orbit.

(NCERT Ed. 2025-26, 11th, Page No. 150)

30. (4)

Drop $h' = \frac{3h}{4}$.

Impact $v = \sqrt{2g(3h/4)} = \sqrt{3gh/2}$.

Elastic rebound, $\Delta v = 2v$.

$J = 2m\sqrt{3gh/2} = m\sqrt{6gh}$.

(NCERT Ed. 2025-26, 11th, Page No. 75)

31. (1)

$$\frac{\mu_0 I_D I_C}{2\pi x} = \frac{\mu_0 I_B I_C}{2\pi(15-x)} \Rightarrow \frac{15}{x} = \frac{10}{15-x} \Rightarrow x = 9 \text{ cm.}$$

(NCERT Ed. 2025-26, 12th, Page No. 123)

32. (2)

Effective length $L = l$. $E = Bvl$.

(NCERT Ed. 2025-26, 12th, Page No. 163)

33. (3)

Accelerated charges produce EM waves.

(NCERT Ed. 2025-26, 12th, Page No. 205)

34. (1)

Density is independent of A because $R \propto A^{1/3}$ and Volume $\propto A$.

(NCERT Ed. 2025-26, 12th, Page No. 309)

35. (2)

Flux depends only on enclosed charge (Gauss Law).

(NCERT Ed. 2025-26, 12th, Page No. 30)

36. (2)

$$P = \frac{VQ}{t} = \frac{5\times10^6\times10}{0.1} = 500 \text{ MW.}$$

(NCERT Ed. 2025-26, 12th, Page No. 92)

37. (4)

$r \to \frac{r}{2} \Rightarrow A \to \frac{A}{4} \Rightarrow L \to 4L$.

$R \propto \frac{L}{A} \Rightarrow R_{new} = 16R = 320\ \Omega$.

(NCERT Ed. 2025-26, 12th, Page No. 84)

38. (1)

$g \propto \rho R$. $\frac{g_p}{g_e} = \frac{\rho_p R_p}{\rho_e R_e} \Rightarrow 2 = 4\frac{R_p}{R} \Rightarrow R_p = \frac{R}{2}$.

(NCERT Ed. 2025-26, 11th, Page No. 146)

39. (3)

$W = T \cdot 2 \cdot \Delta A = T \cdot 8\pi(3^2 - 1^2) \times 10^{-4}$

$\Rightarrow 6.4\pi\times10^{-4} = T \cdot 64\pi\times10^{-4} \Rightarrow T = 0.1 \text{ N m}^{-1}$.

(NCERT Ed. 2025-26, 11th, Page No. 201)

40. (2)

$$\frac{C}{100} = \frac{70-20}{120-20} = \frac{50}{100} \Rightarrow 50^\circ \text{C.}$$

(NCERT Ed. 2025-26, 11th, Page No. 211)

41. (1)

$P \propto R^2T^4$.

$P_{new} \propto (2R)^2\left(\frac{T}{2}\right)^4$

$= \frac{1}{4}P_{old} = 50 \text{ W.}$

(NCERT Ed. 2025-26, 11th, Page No. 224)

42. (3)

$v_{boy} = 2V_{man}$.

$\frac{1}{2}(4m)(V+5)^2 = 4\left(\frac{1}{2}m(2V)^2\right)$

$\Rightarrow (V+5)^2 = 4V^2 \Rightarrow V = 5 \text{ m/s.}$

(NCERT Ed. 2025-26, 11th, Page No. 97)

43. (2)

$W = MB(1-(-1)) = 2\times0.5\times0.2 = 0.2 \text{ J.}$

(NCERT Ed. 2025-26, 12th, Page No. 139)

44. (3)

$r \propto \frac{n^2}{Z}$.

$\frac{n^2}{9} = 1 \Rightarrow n = 3$.

(NCERT Ed. 2025-26, 12th, Page No. 299)

45. (3)

$h_{app} = \mu h_{real} = \frac{4}{3}\times36 = 48 \text{ cm.}$

(NCERT Ed. 2025-26, 12th, Page No. 229)

[Contd...

46. (3)

Rate has one term proportional to [RX] and another to [RX][OH^-], showing parallel unimolecular and bimolecular paths; hence mixed S_N1/S_N2 mechanism operating under given conditions.

(NCERT Ed. 2025-26, 12th, Page No. 179)

47. (4)

Base removes β-hydrogen while halide leaves in one step, giving alkene; more substituted but-2-ene dominates by Zaitsev, so it's dehydrohalogenation a classic β-elimination, not dehydration.

(NCERT Ed. 2025-26, 12th, Page No. 182)

48. (1)

Monocarboxylic acids follow general formula CnH_2nO_2; first is HCOOH (n = 1), second uses n = 2 giving $C_2H_4O_2$ so the second homologue is acetic acid with that formula.

(NCERT Ed. 2025-26, 11th, Page No. 262)

49. (4)

Al_2O_3 and ZnO react with both acids and bases, showing amphoteric nature; Na_2O is basic and CO is neutral, so only that set fits correctly.

(NCERT Ed. 2025-26, 11th, Page No. 94)

50. (2)

From $N_2 + 3H_2 \rightarrow 2NH_3$, 2 mol NH_3 need 3 mol H_2; for 12 mol NH_3, H_2 = 18 mol by simple stoichiometric ratio.

(NCERT Ed. 2025-26, 11th, Page No. 22)

51. (1)

$$Mg^{2+} + NH_3 + HPO_4^{2-} \rightarrow MgNH_4PO_4 \downarrow$$

(NCERT Ed. 2025-26, 11th, Page No. 285)

52. (2)

Double salts and simple salts dissociate completely in water and show tests for all constituent ions, but a complex salt retains its complex ion. Thus $K_4[Fe(CN)_6]$ gives K^+ and $[Fe(CN)_6]^{4-}$, not separate Fe^{2+} and CN^- tests.

(NCERT Ed. 2025-26, 12th, Page No. 120)

53. (3)

$$R-X + AgCN \rightarrow R-NC + AgX$$

(NCERT Ed. 2025-26, 12th, Page No. 176)

54. (1)

Fe^{3+} is $3d^5$ and Mn^{2+} is also $3d^5$. With H_2O as a weak-field ligand, both octahedral complexes are high spin and contain five unpaired electrons each, so their spin-only magnetic moments are equal.

(NCERT Ed. 2025-26, 12th, Page No. 104)

55. (3)

$$\frac{1}{\lambda} = R \cdot (2)^2 \left(\frac{1}{2^2} - \frac{1}{5^2}\right) = R \cdot 4\left(\frac{1}{4} - \frac{1}{25}\right)$$

$$= R \cdot 4\left(\frac{25-4}{100}\right) = R \cdot 4 \cdot \frac{21}{100} = R \cdot \frac{84}{100} = R \cdot \frac{21}{25}$$

$$\lambda = \frac{25}{21R}$$

(NCERT Ed. 2025-26, 11th, Page No. 48)

56. (1)

Calomel electrode consists of mercury, mercurous chloride and chloride ions, so it is metal-insoluble salt-anion type.

(NCERT Ed. 2025-26, 12th, Page No. 35)

57. (1)

Fe^{3+} is $3d^5$ stable and Fe^{2+} tends to oxidise to it, explaining their redox behaviour.

(NCERT Ed. 2025-26, 12th, Page No. 108)

58. (1)

For KCl, van't Hoff factor increases on dilution, so $i_A < i_B < i_C$.

(NCERT Ed. 2025-26, 12th, Page No. 25)

59. (2)

CO has all electrons paired in molecular orbitals, whereas O_2, NO and B_2 each contain unpaired electrons, making them paramagnetic..

(NCERT Ed. 2025-26, 11th, Page No. 131)

60. (1)

Na_2O is basic oxide.

(NCERT Ed. 2022-23, 11th, Page No. 319)

61. (1)

$$H_2C=CH-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}=O$$

(NCERT Ed. 2025-26, 12th, Page No. 209)

[Contd...

62. (1)

Using $\Delta x \Delta p = \frac{h}{4\pi}$

$$\Delta x = \frac{6.626 \times 10^{-34}}{8\pi \times 10^{-24}} \approx 2.64 \times 10^{-11}\ m$$

(NCERT Ed. 2025-26, 11th, Page No. 51)

63. (2)

Entropy increases most when a solid decomposes to produce a gas, greatly increasing disorder.

(NCERT Ed. 2025-26, 11th, Page No. 159)

64. (4)

In (4) lowest sum is (1 + 3 + 6) = 10 and numbering should be from alkene side (alphabetical order) hence correct.

(NCERT Ed. 2025-26, 11th, Page No. 268)

65. (2)

First-order rate is k[A], so rate equals $2 \times 10^{-3} \times 3 = 6 \times 10^{-3}$ mol L^{-1} s^{-1}.

(NCERT Ed. 2025-26, 12th, Page No. 69)

66. (3)

Heat depends on path of process, whereas enthalpy, internal energy, and Gibbs free energy depend only on state.

(NCERT Ed. 2025-26, 11th, Page No. 139)

67. (2)

A thin, adherent Al_2O_3 film protects aluminium surface and prevents further reaction with water.

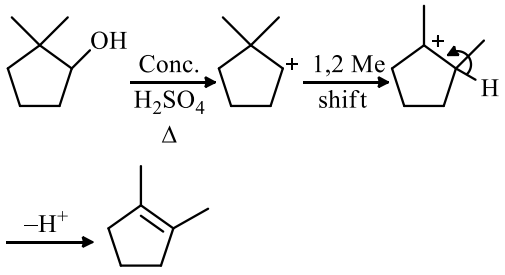
(NCERT Ed. 2022-23, 11th, Page No. 319)

68. (3)

Λ_m	S $cm^2 mol^{-1}$
E_{cell}	V
κ	S cm^{-1}
G*	m^{-1}

(NCERT Ed. 2025-26, 12th, Page No. 45)

69. (2)



(NCERT Ed. 2025-26, 12th, Page No. 211)

70. (4)

Variable oxidation states allow formation of intermediate complexes and easy electron transfer during catalysis.

(NCERT Ed. 2025-26, 12th, Page No. 104)

71. (2)

Moles of water initially $= \frac{90}{18} = 5$

Moles of solute $= \frac{9}{M}$ (let molar mass = M)

$$6 = \frac{5}{5 + \frac{9}{M}} \cdot P^0 \qquad \ldots (1)$$

After adding 1 mole water

Water = 6 mol

$$\frac{36}{39} = \frac{5\left(6 + \frac{9}{M}\right)}{6\left(5 + \frac{9}{M}\right)};\ \frac{12}{13} = \frac{5(6+a)}{6(5+a)} \text{ where } a = \frac{9}{M}$$

$$7a = 30 \Rightarrow a = \frac{30}{7}$$

From (1): $36 = \frac{5}{5 + \frac{30}{7}} P^0$

$$P^0 = \frac{36 \times 65}{35} \approx 66.9\ \text{Torr}$$

(NCERT Ed. 2025-26, 12th, Page No. 12)

72. (4)

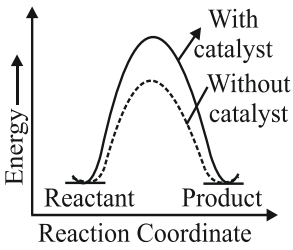
n = 3 (No. of stereo unit)

Meso structure = 0

Total stereoisomers = $2^3 - 0 - 8$

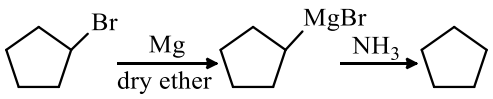
(NCERT Ed. 2025-26, 12th, Page No. 184)

73. (4)



(NCERT Ed. 2025-26, 12th, Page No. 78)

74. (4)



(NCERT Ed. 2025-26, 11th, Page No. 12)

[Contd...

75. (2)

Leading zeros are insignificant, captive zeros count, and trailing decimal zeros count; therefore A, B and C each have three.

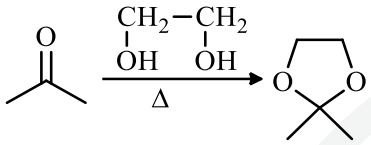
(NCERT Ed. 2025-26, 11th, Page No. 12)

76. (1)

- Benzene: CO, HCl and $AlCl_3$
- Benzonitrile: HCl and $SnCl_2$, H_3O^+
- Benzoyl chloride: H_2, Pd-$BaSO_4$, S and quinoline
- Toluene: CrO_2Cl_2/H^+

(NCERT Ed. 2025-26, 12th, Page No. 233)

77. (2)



(NCERT Ed. 2025-26, 12th, Page No. 237)

78. (2)

–OH is ortho-para directing, –CN meta directing, halides deactivating ortho-para directors, but $–SO_3H$ is meta directing, so D is incorrect.

(NCERT Ed. 2025-26, 11th, Page No. 325)

79. (3)

Primary amines form sulfonamides containing acidic N–H, which dissolve in alkali. Secondary products lack N–H; tertiary amines do not react.

(NCERT Ed. 2025-26, 12th, Page No. 271)

80. (3)

SO_2 and NO_2^- each contain three electron domains around central atom, giving sp^2; only N_3^- is linear with sp hybridization.

(NCERT Ed. 2025-26, 11th, Page No. 121)

81. (2)

Denaturation disrupts higher-order folding like tertiary structure without hydrolyzing peptide bonds. Thus sequence stays same while tertiary structure is lost.

(NCERT Ed. 2025-26, 12th, Page No. 302)

82. (3)

From $N_2 + 3H_2 \rightarrow 2NH_3$, 100 mol N_2 needs 300 mol H_2, so N_2 limits and forms 200 NH_3.

(NCERT Ed. 2025-26, 11th, Page No. 22)

83. (3)

Across period ionization generally rises but Be exceeds B and N exceeds O because filled and half-filled subshells are stable.

(NCERT Ed. 2025-26, 11th, Page No. 89)

84. (3)

Acetone semicarbazone forms by condensation with semicarbazide, $NH_2–NH–CO–NH_2$, which contains three nitrogen atoms; product retains all three nitrogens.

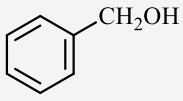
(NCERT Ed. 2025-26, 12th, Page No. 238)

85. (3)

Reactivity does not increase uniformly across a period, but group 1 oxides are basic whereas group 17 oxides are acidic.

(NCERT Ed. 2025-26, 11th, Page No. 94)

86. (4)



(NCERT Ed. 2025-26, 12th, Page No. 238)

87. (2)

Aspirin is acetylsalicylic acid containing one carboxylic acid group, while sulphanilic, picric, and ascorbic acids do not contain –COOH.

(NCERT Ed. 2025-26, 12th, Page No. 210)

88. (1)

Ribose is a D-pentose and free ribose is reducing. In nucleic acids it prefers furanose, not pyranose; α form incorrect.

(NCERT Ed. 2025-26, 12th, Page No. 305)

89. (1)

Statements D and E are incorrect: +5 stability decreases down group, and nitrogen cannot exceed covalency four without d-orbitals.

(NCERT Ed. 2022-23, 12th, Page No. 173)

90. (2)

BrF_5 and $XeOF_4$ each have six electron domains around central atom, so electron-pair geometry is octahedral though molecular shape differs.

(NCERT Ed. 2022-23, 12th, Page No. 210)

[Contd...

91. (2)

The waste water treatment is carried out in two stages:

Primary treatment : These treatment steps basically involve physical removal of particles – large and small – from the sewage through filtration and sedimentation. These are removed in stages; initially, floating debris is removed by sequential filtration. Then the grit (soil and small pebbles) are removed by sedimentation. All solids that settle form the primary sludge, and the supernatant forms the effluent. The effluent from the primary settling tank is taken for secondary treatment.

Secondary treatment or Biological treatment : The primary effluent is passed into large aeration tanks where it is constantly agitated mechanically and air is pumped into it. This allows vigorous growth of useful aerobic microbes into flocs (masses of bacteria associated with fungal filaments to form mesh like structures). While growing, these microbes consume the major part of the organic matter in the effluent. This significantly reduces the BOD (biochemical oxygen demand) of the effluent. BOD refers to the amount of the oxygen that would be consumed if all the organic matter in one liter of water were oxidised by bacteria. The sewage water is treated till the BOD is reduced. The BOD test measures the rate of uptake of oxygen by micro-organisms in a sample of water and thus, indirectly, BOD is a measure of the organic matter present in the water. The greater the BOD of waste water, more is its polluting potential. Once the BOD of sewage or waste water is reduced significantly, the effluent is then passed into a settling tank where the bacterial 'flocs' are allowed to sediment. This sediment is called activated sludge. A small part of the activated sludge is pumped back into the aeration tank to serve as the inoculum. The remaining major part of the sludge is pumped into large tanks called anaerobic sludge digesters. Here, other kinds of bacteria, which grow anaerobically, digest the bacteria and the fungi in the sludge. During this digestion, bacteria produce a mixture of gases such as methane, hydrogen sulphide and carbon dioxide. These gases form biogas and can be used as source of energy as it is inflammable. The effluent from the secondary treatment plant is generally released into natural water bodies like rivers and streams.

(NCERT Ed. 2025-26, 12th, Page No. 154)

92. (3)

Fatty acids would be broken down to acetyl CoA before entering the respiratory pathway when it is used as a substrate. But when the organism needs to synthesise fatty acids, acetyl CoA would be withdrawn from the respiratory pathway for it.

(NCERT Ed. 2025-26, 11th, Page No. 162)

93. (1)

Characteristics of the Five Kingdoms

Characters	Five Kingdoms				
	Monera	Protista	Fungi	Plantae	Animalia
Cell type	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Cell wall	Noncellulosic (Polysaccharide + amino acid)	Present in some	Present with chitin	Present (cellulose)	Absent
Nuclear membrane	Absent	Present	Present	Present	Present
Body organisation	Cellular	Cellular	Multicellular / loose tissue	Tissue / organ	Tissue / organ / organ system
Mode of nutrition	Autotrophic (chemosynthetic and photosynthetic) and Heterotrophic (saprophytic / parasitic)	Autotrophic (Photosynthetic) and Heterotrophic	Heterotrophic (Saprophytic / Parasitic)	Autotrophic (Photosynthetic)	Heterotrophic (Holozoic / Saprophytic etc.)

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94. (2)

There are instances where a single gene can exhibit multiple phenotypic expression. Such a gene is called a pleiotropic gene. The underlying mechanism of pleiotropy in most cases is the effect of a gene on metabolic pathways which contribute towards different phenotypes. An example of this is the disease phenylketonuria, which occurs in humans.

(NCERT Ed. 2025-26, 12th, Page No. 69)

95. (4)

Though the seeds differ greatly, the early stages of embryo development (embryogeny) are similar in both monocotyledons and dicotyledons.

(NCERT Ed. 2025-26, 12th, Page No. 19)

96. (3)

A population growing in a habitat with limited resources show initially a lag phase, followed by phases of acceleration and deceleration and finally an asymptote, when the population density reaches the carrying capacity.

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[Contd...

97. (2)

James Dewey Watson received his Ph.D. degree in 1950 on a study of the effect of hard X-rays on bacteriophage multiplication.

(NCERT Ed. 2025-26, 12th, Page No. 52)

98. (1)

In the seeds of cereals such as maize the seed coat is membranous and generally fused with the fruit wall. The endosperm is bulky and stores food. The outer covering of endosperm separates the embryo by a proteinous layer called aleurone layer.

(NCERT Ed. 2025-26, 11th, Page No. 67)

99. (4)

Dinoflagellates: The cell wall has stiff cellulose plates on the outer surface.

(NCERT Ed. 2025-26, 11th, Page No. 15)

100. (3)

In case of honey bees, the females are diploid having 32 chromosomes and males are haploid, i.e., having 16 chromosomes.

(NCERT Ed. 2025-26, 12th, Page No. 71)

101. (4)

One single maize root apical meristem can give rise to more than 17,500 new cells per hour, whereas cells in a watermelon may increase in size by upto 3,50,000 times. In the former, growth is expressed as increase in cell number; the latter expresses growth as increase in size of the cell. While the growth of a pollen tube is measured in terms of its length, an increase in surface area denotes the growth in a dorsiventral leaf.

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102. (1)

During cell cycle, DNA replicates in S phase of interphase (one of the sub-stages of the resting phase). M phase is the most dramatic phase of cell cycle. For convenience mitosis has been divided into four stages of nuclear division (karyokinesis). DNA does not replicate in prophase I of meiosis.

(NCERT Ed. 2025-26, 11th, Page No. 121)

103. (3)

Light reactions or the 'Photochemical' phase include light absorption, water splitting, oxygen release, and the formation of high-energy chemical intermediates, ATP and NADPH.

(NCERT Ed. 2025-26, 11th, Page No. 138)

104. (2)

The *lac* operon consists of one regulatory gene and three structural genes (*z*, *y*, and *a*). The *i* gene codes for the repressor of the *lac* operon. The *z* gene codes for beta-galactosidase (β-gal), which is primarily responsible for the hydrolysis of the disaccharide, lactose into its monomeric units, galactose and glucose. The y gene codes for permease, which increases permeability of the cell to β-galactosides. The *a* gene encodes a transacetylase.

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105. (2)

Each category, referred to as a unit of classification, in fact, represents a rank and is commonly termed as taxon (pl.: taxa). The species is considered the basic or fundamental unit of classification. An organism belongs to multiple taxonomic categories simultaneously. A higher category contains far more organisms than a lower category.

(NCERT Ed. 2025-26, 11th, Page No. 6)

106. (3)

Brood parasitism in birds is a fascinating example of parasitism in which the parasitic bird lays its eggs in the nest of its host and lets the host incubate them.

(NCERT Ed. 2025-26, 12th, Page No. 200)

107. (2)

Griffith, in his experiments, observed transformation of a non-virulent strain (R strain) to virulent strain (S strain) of pneumococcus. Francis Crick proposed the Central dogma in molecular biology, which states that the genetic information flows from DNA→RNA→Protein.

(NCERT Ed. 2025-26, 12th, Page No. 85)

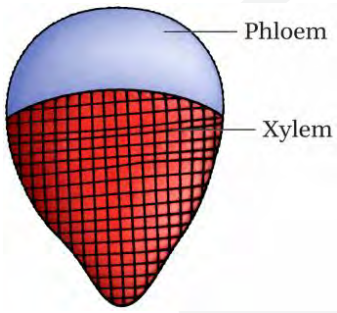
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108. (2)

Identification and enumeration of plant and animal species of an ecosystem gives its species composition. Vertical distribution of different species occupying different levels is called stratification. The components of the ecosystem are seen to function as a unit when you consider the following aspects: (i) Productivity; (ii) Decomposition; (iii) Energy flow; and (iv) Nutrient cycling. The decomposers are the fungi, bacteria and flagellates especially abundant in the bottom of the pond. There is unidirectional movement of energy towards the higher trophic levels and its dissipation and loss as heat to the environment.

(NCERT Ed. 2025-26, 12th, Page No. 206)

109. (1)



Conjoint closed vascular bundle

(NCERT Ed. 2025-26, 11th, Page No. 73)

110. (4)

Telophase I: The nuclear membrane and nucleolus reappear, cytokinesis follows and this is called as dyad of cells.

(NCERT Ed. 2025-26, 11th, Page No. 127)

111. (3)

When the placenta is axial and the ovules are attached to it in a multilocular ovary, the placentation is said to be axile, as in china rose (family Malvaceae). In parietal placentation, the ovules develop on the inner wall of the ovary or on peripheral part. Ovary is one-chambered but it becomes two-chambered due to the formation of the false septum, e.g., mustard and *Argemone*.

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112. (2)

The stroma of the chloroplast contains enzymes required for the synthesis of carbohydrates and proteins. It also contains small, double-stranded circular DNA molecules and ribosomes. Chlorophyll pigments are present in the thylakoids.

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113. (1)

Some plants such as *Viola* (common pansy), *Oxalis*, and *Commelina* produce two types of flowers – chasmogamous flowers which are similar to flowers of other species with exposed anthers and stigma, and cleistogamous flowers which do not open at all. Papaya is a dioecious plant, meaning male and female flowers are typically produced on separate plants. Its flowers are chasmogamous (open) to allow for cross-pollination by insects or wind. It does not produce cleistogamous flowers.

(NCERT Ed. 2025-26, 12th, Page No. 12)

114. (4)

Alien species invasions: When alien species are introduced unintentionally or deliberately for whatever purpose, some of them turn invasive, and cause decline or extinction of indigenous species. The recent illegal introduction of the African catfish *Clarias gariepinus* for aquaculture purposes is posing a threat to the indigenous catfishes in our rivers.

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115. (3)

Gibberellins promotes bolting (internode elongation just prior to flowering) in beet, cabbages and many plants with rosette habit. Apical dominance and phototropism are effects of auxins. Ethylene is highly effective in fruit ripening. It enhances the respiration rate during ripening of the fruits. This rise in rate of respiration is called respiratory climactic.

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116. (2)

The gametes can be flagellated and similar in size (as in *Ulothrix*) or non-flagellated (non-motile) but similar in size (as in *Spirogyra*). Such reproduction is called isogamous. Fusion between one large, non-motile (static) female gamete and a smaller, motile male gamete is termed oogamous, e.g., *Volvox*, *Fucus* and *Gelidium*.

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[Contd...

117. (2)

While each of the membranous organelles is distinct in terms of its structure and function, many of these are considered together as an endomembrane system because their functions are coordinated. The endomembrane system include endoplasmic reticulum (ER), golgi complex, lysosomes and vacuoles.

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118. (1)

The primary transcripts contain both the exons and the introns and are non-functional. hnRNA undergoes additional processing called as capping and tailing. In capping an unusual nucleotide (methyl guanosine triphosphate) is added to the 5'-end of hnRNA. It is subjected to a process called splicing where the introns are removed and exons are joined in a defined order. In tailing, adenylate residues (200-300) are added at 3'-end in a template independent manner. It is the fully processed hnRNA, now called mRNA, that is transported out of the nucleus for translation.

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119. (3)

Crossing over is the exchange of genetic material between two homologous chromosomes. Crossing over is also an enzyme-mediated process and the enzyme involved is called recombinase. Crossing over leads to recombination of genetic material between non-sister chromatids of the two homologous chromosomes.

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120. (2)

Calvin proposed that plants change light energy to chemical energy by transferring an electron in an organised array of pigment molecules and other substances. The mapping of the pathway of carbon assimilation in photosynthesis earned him Nobel Prize in 1961.

(NCERT Ed. 2025-26, 11th, Page No. 132)

121. (3)

Streptokinase produced by the bacterium *Streptococcus* and modified by genetic engineering is used as a 'clot buster' for removing clots from the blood vessels. Another bioactive molecule, cyclosporin A, that is used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus *Trichoderma polysporum*. Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents. Lipases are used in detergent formulations and are helpful in removing oily stains from the laundry.

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122. (4)

There is a characteristic distribution of the cells within the embryo sac. Three cells are grouped together at the micropylar end and constitute the egg apparatus. The egg apparatus, in turn, consists of two synergids and one egg cell. The synergids have special cellular thickenings at the micropylar tip called filiform apparatus, which play an important role in guiding the pollen tubes into the synergid. Three cells are at the chalazal end and are called the antipodals. The large central cell has two polar nuclei. Thus, a typical angiosperm embryo sac, at maturity, though 8-nucleate is 7-celled.

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123. (1)

In an isobilateral leaf, the stomata are present on both the surfaces of the epidermis; and the mesophyll is not differentiated into palisade and spongy parenchyma. In grasses, certain adaxial epidermal cells along the veins modify themselves into large, empty, colourless cells. These are called bulliform cells. The parallel venation in monocot leaves is reflected in the near similar sizes of vascular bundles (except in main veins).

(NCERT Ed. 2025-26, 11th, Page No. 77)

124. (3)

The given cross leads to a phenotypic ratio of 3/4th tall : (1/4 TT + 1/2 Tt) and 1/4th tt, i.e., a 3:1 ratio, but a genotypic ratio of 1:2:1, in the F_2 generation.

(NCERT Ed. 2025-26, 12th, Page No. 58)

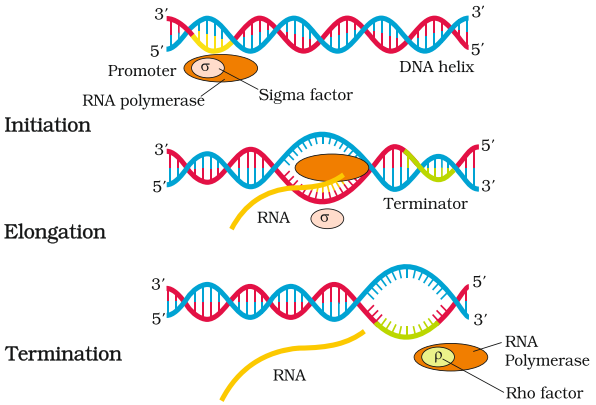
125. (2)

Marchantia is a liverwort. *Polytrichum* is a moss.

(NCERT Ed. 2025-26, 11th, Page No. 30)

[Contd...

126. (3)



Process of Transcription in Bacteria

(NCERT Ed. 2025-26, 12th, Page No. 93)

127. (4)

In modern medicine certain infectious neurological diseases were found to be transmitted by an agent consisting of abnormally folded protein. The agent was similar in size to viruses. These agents were called prions. The most notable diseases caused by prions are bovine spongiform encephalopathy (BSE) commonly called mad cow disease in cattle and its analogous variant Creutzfeldt-Jacob disease (CJD) in humans.

(NCERT Ed. 2025-26, 11th, Page No. 21)

128. (4)

In fermentation, say by yeast, the incomplete oxidation of glucose is achieved under anaerobic conditions by sets of reactions where pyruvic acid is converted to CO_2 and ethanol. The enzymes, pyruvic acid decarboxylase and alcohol dehydrogenase catalyse these reactions. Other organisms like some bacteria produce lactic acid from pyruvic acid. In animal cells also, like muscles during exercise, when oxygen is inadequate for cellular respiration pyruvic acid is reduced to lactic acid by lactate dehydrogenase.

(NCERT Ed. 2025-26, 11th, Page No. 157)

129. (1)

The parenchymatous cells which lie between the xylem and the phloem are called conjuctive tissue.

(NCERT Ed. 2025-26, 11th, Page No. 74)

130. (3)

The whole scheme of transfer of electrons, starting from the PS II, uphill to the acceptor, down the electron transport chain to PS I, excitation of electrons, transfer to another acceptor, and finally down hill to $NADP^+$ reducing it to $NADPH + H^+$ is called the Z scheme.

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131. (2)

Microvilli are microscopic, finger-like protrusions of the plasma membrane. They are most famously found in the animal small intestine, where they are essential for efficiently absorbing nutrients from digested food into the bloodstream.

(NCERT Ed. 2025-26, 11th, Page No. 92)

132. (1)

According to Robert May's global estimates, only 22 per cent of the total species have been recorded so far.

(NCERT Ed. 2025-26, 12th, Page No. 219)

133. (3)

In an aquatic ecosystem, GFC is the major conduit for energy flow. As against this, in a terrestrial ecosystem, a much larger fraction of energy flows through the detritus food chain than through the GFC.

(NCERT Ed. 2025-26, 12th, Page No. 210)

134. (1)

In some plants such as *Rhizophora* growing in swampy areas, many roots come out of the ground and grow vertically upwards. Such roots, called pneumatophores, help to get oxygen for respiration. Hanging structures that support a banyan tree are called prop roots. The stems of maize and sugarcane have supporting roots coming out of the lower nodes of the stem. These are called stilt roots. Underground stems of potato, ginger, turmeric are modified to store food in them.

(NCERT Ed. 2025-26, 11th, Page No. 58)

135. (4)

Certain marine brown and red algae produce large amounts of hydrocolloids (water holding substances), e.g., algin (brown algae) and carrageen (red algae) which are used commercially.

(NCERT Ed. 2025-26, 11th, Page No. 26)

[Contd...

136. (2)

Ammonia is highly toxic and requires large amounts of water for elimination. Uricotelic animals like birds and reptiles conserve water by excreting nitrogenous waste as a paste or pellet. Mammals are ureotelic and excrete urea, while ammonia excretion through the body surface is typical of aquatic animals like fishes. In aquatic organisms, ammonia is primarily lost by diffusion across gill surfaces or body surfaces, and kidneys do not play a significant role in its removal.

(NCERT Ed. 2025-26, 11th, Page No. 42)

137. (3)

The demographic status of a country is heavily influenced by the balance between birth and death rates. If the Maternal Mortality Rate (MMR) and Infant Mortality Rate (IMR) decline, it means more mothers and infants are surviving. If the birth rate remains high and unchanged during this period, the number of individuals added to the population far exceeds the number of deaths, leading to rapid population expansion.

(NCERT Ed. 2025-26, 11th, Page No. 110)

138. (3)

List-I	List-II
Opioids	Bind to specific opioid receptors present in our central nervous system and gastrointestinal tract
Cannabinoids	Obtained from the inflorescences of the plant *Cannabis sativa*
Cocaine	Has a potent stimulating action on central nervous system
Barbiturates	Normally used as medicines to help patients cope with mental illnesses

(NCERT Ed. 2025-26, 11th, Page No. 45)

139. (4)

Malignant tumors are characterized by rapidly dividing neoplastic cells that have lost the property of "contact inhibition". These cells have a very high metabolic demand. They indirectly cause the death of nearby normal cells by competing for vital nutrients, essentially starving the healthy tissues to support their own uncontrolled growth.

(NCERT Ed. 2025-26, 12th, Page No. 168)

140. (3)

Living organisms and the non-living world (like the earth's crust) are made of the same basic elements. However, the proportion or "relative abundance" of these elements differs significantly. Specifically, the relative abundance of carbon and hydrogen is much higher in any living organism compared to the earth's crust.

(NCERT Ed. 2025-26, 11th, Page No. 201)

141. (3)

The cerebral cortex contains specific functional areas. The motor area controls voluntary movements, and the sensory area processes incoming stimuli. However, there are large regions that are neither clearly sensory nor motor; these are called "association areas". These areas are responsible for complex functions like memory, communication, and intersensory associations.

(NCERT Ed. 2025-26, 11th, Page No. 212)

142. (3)

Recombinant therapeutics (like human insulin) are safer as they do not produce unwanted immunological responses.

PCR (Polymerase Chain Reaction) is a powerful tool used to detect low levels of pathogens or specific genetic mutations in suspected cancer patients.

(NCERT Ed. 2025-26, 11th, Page No. 222)

143. (3)

A "true coelom" is a body cavity completely lined by mesoderm. Aschelminthes are pseudocoelomates, meaning their cavity is not fully lined by mesoderm. Platyhelminthes are acoelomates (no cavity), and Porifera, Coelenterates, and Ctenophores lack a coelom entirely. The group that includes only true coelomates is Arthropods, Hemichordates, and Echinoderms (along with Annelids, Molluscs, and Chordates).

(NCERT Ed. 2025-26, 11th, Page No. 235)

144. (2)

Gout is a specific metabolic disorder. It is characterized by the inflammation of joints, which occurs due to the abnormal accumulation of uric acid crystals in the joint space.

(NCERT Ed. 2025-26, 11th, Page No. 243)

145. (1)

Chorionic villi are foetal tissues, while the uterine tissue is maternal. The placenta is a structural and functional unit formed by the interdigitation of these two tissues. The placenta is a vital endocrine organ that secretes hormones like estrogens and progesterone. hCG and hPL are hormones produced only during pregnancy to support the fetus.

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[Contd...

146. (4)

In a glycogen chain, the right end is the reducing end and the left is non-reducing. Cellulose is a homopolymer of glucose only, not a heteropolymer. Unlike starch, cellulose does not form complex helices and therefore cannot hold iodine or turn blue. Chitin is a complex polysaccharide found in insect exoskeletons and is a homopolymer of N-acetyl glucosamine.

(NCERT Ed. 2025-26, 12th, Page No. 46)

147. (3)

The PCT reabsorbs nearly 70-80% of electrolytes and water, not just 10%. The descending limb is permeable to water but almost impermeable to electrolytes. The ascending limb is impermeable to water but allows electrolyte transport, making the filtrate hypotonic, not hypertonic. The DCT is the site where "conditional" reabsorption of sodium and water occurs, often under the influence of hormones.

(NCERT Ed. 2025-26, 12th, Page No. 115)

148. (1)

List-I	List-II
Pila	File-like rasping organ called radula for feeding.
Nereis	Dioecious aquatic form with lateral appendages called parapodia.
Carcharodon	Marine fish with ventral mouth
Pleurobrachia	Bioluminescent organism with external fertilisation

(NCERT Ed. 2025-26, 12th, Page No. 133)

149. (3)

Biopiracy is a term used to describe the unethical use of biological resources. It refers specifically to the exploitation of bio-resources by multi-national companies or other organizations without proper authorization from the countries or people concerned, and without providing fair compensation.

(NCERT Ed. 2025-26, 12th, Page No. 170)

150. (2)

Medical consultation is necessary because contraceptives are not natural requirements for the body and can have side effects. While they are effective at preventing pregnancy, they are not universally free from side effects; they can cause nausea, abdominal pain, or irregular bleeding. They do not permanently destroy ovarian reserve or guarantee protection from all reproductive disorders, so professional advice ensures the best method is chosen for an individual's health.

(NCERT Ed. 2025-26, 12th, Page No. 185)

151. (2)

Emphysema is a chronic respiratory disorder often caused by cigarette smoking. It is specifically characterized by the damage and breakdown of the alveolar walls. This leads to a significant decrease in the total surface area available for gas exchange.

(NCERT Ed. 2025-26, 11th, Page No. 46)

152. (1)

Rheumatoid arthritis is an autoimmune disease where the body's immune system attacks its own joints.

Auto-immunity is not "exclusively" due to genetic mutations; it is a complex result of both genetic and environmental factors where the body loses the ability to distinguish "self" from "non-self".

(NCERT Ed. 2025-26, 11th, Page No. 106)

153. (2)

The plasmid pBR322 has two primary antibiotic resistance genes: amp^R and tet^R. The recognition site for the restriction enzyme *Bam* HI is located within the tetracycline resistance gene. If foreign DNA is inserted at this specific site, it disrupts the gene sequence—a process called insertional inactivation—resulting in the loss of tetracycline resistance only. Ampicillin resistance remains unaffected.

(NCERT Ed. 2020-21, 11th, Page No. 112)

154. (2)

Blood clotting is a cascade of enzymatic reactions. The first major step involves the formation of the enzyme complex called Thrombokinase. Thrombokinase then catalyzes the conversion of the inactive plasma protein Prothrombin into the active enzyme Thrombin. Thrombin then goes on to convert soluble fibrinogen into insoluble fibrin.

(NCERT Ed. 2025-26, 11th, Page No. 189)

155. (2)

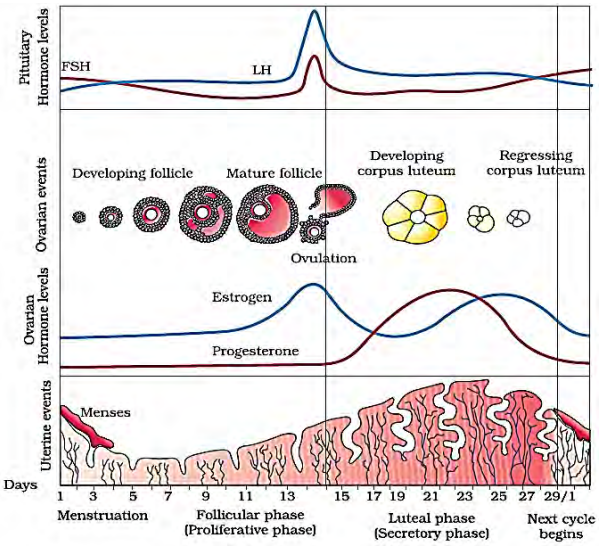
Secondary metabolites are chemicals produced by plants/microbes that aren't part of primary metabolism. Concanavalin A is a Lectin, not an alkaloid. Ricin and Abrin are categorized as Toxins. Codeine is an Alkaloid, not an essential oil. Anthocyanins are Pigments, whereas drugs would include substances like Vinblastine.

(NCERT Ed. 2025-26, 11th, Page No. 200)

[Contd...

156. (4)

In the provided diagram, Stage D represents the end of the cycle where "Menses" (menstruation) begins again.



Menstruation is triggered by the breakdown of the endometrium, which occurs when the corpus luteum degenerates. This degeneration leads to a sudden and sharp decline in the levels of both progesterone and estrogen.

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157. (2)

In male frogs, the anatomy of the reproductive and excretory systems is closely linked. Ureters act as urinogenital ducts but open into the cloaca, not directly into the bladder. Vasa efferentia (10-12 in number) enter the kidney and open into Bidder's canal. Testes are held to the upper part of the kidneys and the copulatory pads are on the first digit of the forelimbs, not hind limbs.

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158. (3)

List-I	List-II
Parathyroid hormone (PTH)	Increases blood calcium level
Calcitonin	Decreases blood calcium level
Insulin	Decreases blood glucose
Aldosterone	Promotes sodium reabsorption in kidney

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159. (4)

RNA interference (RNAi) is a cellular defense mechanism in all eukaryotes that involves silencing specific mRNA to prevent its translation into protein. In biotechnology, both sense and antisense RNA are produced in the host to form double-stranded RNA (dsRNA). dsRNA does not destroy the nematode's DNA; instead, it triggers the degradation of the nematode's mRNA.

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160. (3)

List-I	List-II
Exonuclease	Removes nucleotides from the ends of DNA
Endonuclease	Makes cuts at specific internal positions within DNA
DNA ligase	Joins DNA fragments by forming phosphodiester bonds
Origin of replication	Specific sequence from where replication starts

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161. (4)

In *Periplaneta americana* (cockroach), the 7th sternum is boat-shaped in females, not males. Anal styles are found only in males, not females. Tegmina are the leathery forewings that cover the hindwings; they are opaque, not transparent, and are not primarily used for flight. Mushroom-shaped accessory reproductive glands are indeed located in segments 6-7 of the male abdomen.

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162. (1)

The diffusion membrane is the interface where oxygen and carbon dioxide are exchanged between the air in the alveoli and the blood. It consists of three major layers: the thin squamous epithelium of the alveoli, the endothelium of the surrounding alveolar capillaries, and a basement substance (composed of basement membranes) sandwiched between them.

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163. (4)

List-I (Part)	List-II (No. of bones)
Skull (including hyoid and ear ossicles)	29
Vertebral column	26
Ribs	24
Hind limb (one leg)	30

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164. (2)

Sweet potato (root modification) and potato (stem modification) perform the same function but have different origins, making them analogous. Homology shows common ancestry, while analogy results from convergent evolution where different structures evolve for the same function. The eyes of octopuse and mammal are examples of analogy (different structures, same function), not homology.

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165. (3)

Genetic engineering involves the direct manipulation of DNA. Downstream processing refers to the separation and purification of the product after the process is complete. Bioprocess engineering is the specific term for maintaining sterile conditions (free of contamination) in chemical engineering processes to allow the growth of only the desired microbe or cell in large quantities. Gene cloning is the process of making multiple copies of a gene.

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166. (2)

List-I	List-II
Crop	Sac-like structure used for storing food
Gizzard	Outer thick circular muscles with six chitinous teeth
Hepatic caeca	Ring of 6-8 blind tubules secreting digestive juice
Malpighian tubule	100-150 yellow filamentous tubules for excretion

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167. (2)

GIFT (Gamete Intra-Fallopian Transfer) involves transferring unfertilized eggs and sperm directly into the fallopian tube so fertilization can occur inside the body *(in vivo*). ZIFT (Zygote Intra-Fallopian Transfer) involves transferring a zygote that was fertilized outside (*in vitro*). Because GIFT requires fertilization to happen naturally in the tube, having at least one healthy, patent (open) fallopian tube is absolutely essential. ZIFT also uses the tube, but GIFT cannot function without the tube's natural environment for the actual meeting of gametes.

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168. (4)

The image shows a comparison between a damaged and a healthy cotton boll. This is a classic demonstration of the damage caused by "bollworms". This specific example is often used when discussing Bt Cotton, which was genetically engineered to be resistant to bollworm infestation.

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169. (3)

Sperm count depends on spermatogenesis in the tubules, and release depends on spermiation. However, the maturation and motility of sperm are heavily dependent on the secretions they receive as they pass through the male duct system. Specifically, the secretions from the epididymis, vas deferens, seminal vesicles, and prostate gland are essential for sperm to become fully motile and fertile. Therefore, low motility is most likely due to dysfunctional secretions from these accessory structures.

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170. (3)

Apoenzyme is the protein portion of the enzyme, it is not the non-protein part. Haem is a prosthetic group (tightly bound), not a co-enzyme (loosely bound). NAD and NADP contain the vitamin Niacin, not Riboflavin. Removing a co-factor always destroys catalytic activity.

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171. (1)

In vertebrate embryos, much of the skeleton is made of cartilage, which is later replaced by bone as they grow into adults.

Gap junctions are specifically designed to facilitate communication between cells by allowing the rapid transfer of ions, small molecules, and sometimes even large molecules.

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172. (4)

Adult echinoderms are radial, and larvae are bilateral. Ctenophores reproduce *only* sexually, not asexually. Aschelminthes have a very well-developed muscular pharynx. Sponges are generally asymmetrical, whereas coelenterates have radial symmetry, meaning any plane through the center divides them into equal halves.

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173. (2)

Reptiles do have dry, cornified skin with scales, and it is also true that while most have a three-chambered heart, crocodiles are an exception with four chambers. However, having a four-chambered heart in crocodiles is a circulatory feature and does not "explain" why the skin is dry and scaly. Therefore, both are true, but R is not the correct explanation of A.

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174. (3)

In frogs, the heart is three-chambered (two atria, one ventricle), not four-chambered. The renal portal system connects the kidneys with the lower parts of the body, while the hepatic portal system connects the liver and intestine. The sinus venosus joins the right atrium. Frog red blood cells (erythrocytes) are nucleated and contain haemoglobin.

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175. (2)

During pregnancy, the levels of several hormones in the maternal blood increase many-fold. These include estrogens, progestogens, cortisol, prolactin, and thyroxine. These increases are essential for supporting fetal growth, maintaining the pregnancy, and managing the mother's metabolic changes.

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176. (2)

The heart sounds heard through a stethoscope are associated with the mechanical events of the cardiac cycle. The first sound, 'Lub', is produced by the closure of the tricuspid and bicuspid (AV) valves at the start of ventricular systole. The second sound, 'Dub', is produced by the closure of the semilunar valves at the start of ventricular diastole. They are caused by valves closing (to prevent backflow), not by muscle contraction or valves opening.

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177. (2)

List-I	List-II
Louis Pasteur	Demonstrated life comes from pre-existing life
Oparin & Haldane	Proposed chemical evolution preceded life
S.L. Miller	Created electric discharge in a closed flask at 800 ^{o}C
Hugo de Vries	Proposed mutation as the cause of speciation

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178. (3)

Modern medicine has used vaccines to tackle many infectious diseases. Smallpox has been completely eradicated. Others, like polio, diphtheria, pneumonia, and tetanus, have been controlled to a large extent through successful immunization programs.

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179. (4)

PCR involves a cycle of three steps: denaturation, annealing, and extension. During denaturation, the DNA is heated to very high temperatures (around) to separate the strands. Most enzymes would be destroyed by this heat. *Taq* polymerase, isolated from the bacterium *Thermus aquaticus*, is used because it is "thermostable," meaning it remains active and functional even after the high-temperature denaturation step.

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180. (2)

In an acute stress situation (the "fight or flight" response), the adrenal medulla releases catecholamines (adrenaline and noradrenaline). These hormones increase alertness, dilate the pupils, and increase the heart rate and strength of heart contraction. They also stimulate the breakdown of glycogen into glucose (glycogenolysis) to provide immediate energy, which is the opposite of glycogenesis or protein synthesis.

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